

A 1.1.9 Cloud Computing



A 1.1.9 Describe the different types of services in cloud computing

Most of the software you used this week probably ran on someone else's computer. Cloud computing means renting computing resources over the internet, paying for what you use rather than buying and maintaining hardware yourself. The syllabus names three service models: Software as a Service (SaaS), Platform as a Service (PaaS) and Infrastructure as a Service (IaaS), which differ in how much of the system the provider manages and how much the customer does. This page describes each model, explains how their different degrees of control and flexibility affect resource management and resource availability, and shows how the choice plays out in real-world scenarios.

Key Questions

- What does each of the three cloud service models (SaaS, PaaS, IaaS) provide, and who manages what in each?
- How do the different degrees of control and flexibility influence resource management and resource availability?
- Given a real-world scenario, which model fits, and why?

IB Command Term note

The command term is **describe**: give a detailed account.

A describe question on this topic expects you to state what a service model provides, identify who manages which parts of the system, and say what degree of control the customer keeps. Expect scenario-based questions: given a situation, describe the model in use or the model that fits it. An answer that names a model without describing its characteristics cannot reach full marks.

THINK FIRST

Think of three applications you used this week through a browser or an app: perhaps Gmail, a shared document, a streaming service. Whose computers did the processing? Who installs the updates? Who pays when a hard drive fails? Keep your answers in mind as you read.

What is cloud computing?

Before the cloud, an organisation that wanted to run software bought servers, installed them in a room on site, and employed people to maintain them. Cloud computing replaces that ownership with rental: a provider such as Amazon, Google or Microsoft runs vast data centres, and customers pay for slices of that capacity on demand, over the internet. The question every customer then faces is how much of the system they want to manage themselves. The three service models are three different answers to that question.

The three service models

The models sit on a scale from least control to most. Each step along the scale hands the customer more of the system to manage, and with that responsibility comes more flexibility.

Software as a Service (SaaS)

What it is. SaaS delivers ready-to-use applications over the internet. You open the software in a browser or app and start working: there is nothing to install and no infrastructure to manage. Common examples are Google Workspace, Microsoft 365, Salesforce and Dropbox.

Who manages what. The provider manages everything: the application, the operating system, the servers and the storage. The customer controls only how the software is used and a limited set of settings. This is the lowest degree of control and flexibility: you cannot modify the software itself, and you depend entirely on the provider to manage resources and keep the service available.

A school adopts Google Workspace for email and documents. Staff and students work in a browser; nobody at the school patches a server or worries about storage filling up, but nobody can change how Docs itself behaves either.

Platform as a Service (PaaS)

What it is. PaaS provides a ready-made platform for developing and deploying applications. The customer writes and uploads the code; the provider supplies and manages the runtime, the

operating system and all the hardware underneath. Examples include AWS Elastic Beanstalk, Google App Engine and Microsoft Azure App Service.

Who manages what. The customer controls the application and its configuration, which gives more flexibility than SaaS: you can build whatever software you like, within the languages and services the platform supports. Resource management is largely automatic, because the platform allocates capacity for you, and availability remains the provider's responsibility.

A two-person team builds a revision-tracking web app and deploys it to Google App Engine. They push code; the platform allocates servers, and when traffic spikes on results day it adds capacity automatically.

Infrastructure as a Service (IaaS)

What it is. IaaS rents out fundamental computing resources: virtual servers, storage and networking. The customer builds everything above that layer, starting with the choice of operating system. Examples include Amazon EC2, Microsoft Azure Virtual Machines and Google Compute Engine.

Who manages what. This is the highest degree of control and flexibility: you choose the operating system, install any software and design the whole environment. The price is responsibility. The customer manages the operating system, security patches, software updates and capacity planning. Resource availability is in your hands too: you decide how many servers to rent and how to configure redundancy, so an outage caused by under-provisioning is your failure, not the provider's.

A streaming company rents hundreds of virtual machines on Amazon EC2 and builds a custom video-processing pipeline on them. Its engineers tune every layer to squeeze out performance, and the same engineers carry the responsibility when a server needs patching at three in the morning.

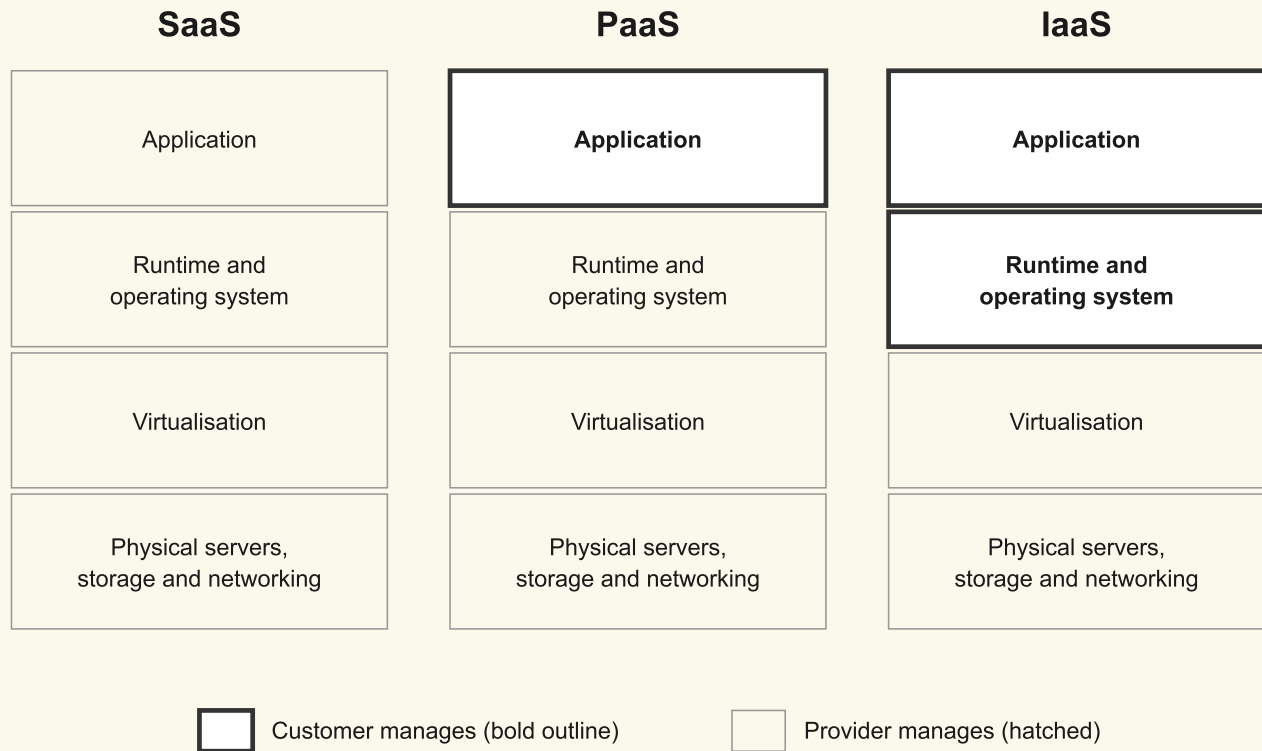


Figure 1. Who manages what in each cloud service model. Moving from SaaS to IaaS transfers layers of the stack from the provider to the customer.

How control and flexibility affect resources

Control and flexibility move together, and they decide two practical questions: who manages the resources, and who is responsible for keeping them available. With SaaS, the provider makes every resource decision and guarantees availability; the customer simply trusts the service. With IaaS, the customer makes those decisions: servers can be tuned precisely to the workload, which is the flexibility, but capacity must be planned and redundancy configured by the customer's own staff. PaaS sits between the two: the customer shapes the application while the platform manages capacity within the limits it supports. The right choice therefore depends on how much control an organisation actually needs and how much technical expertise it can afford.

Feature	SaaS	PaaS	IaaS
Control	Lowest: you control how the application is used and some settings	Medium: you control the application and its configuration	Highest: you control everything from the operating system up

Feature	SaaS	PaaS	IaaS
Flexibility	Lowest: the software cannot be modified	Medium: build any application the platform supports	Highest: design the entire environment
Resource management	Entirely the provider's job	Mostly automatic: the platform allocates capacity	The customer's job: patching, security, capacity planning
Resource availability	Guaranteed by the provider	Handled by the platform, which typically scales automatically	Configured by the customer: under-provisioning is the customer's risk

COMMON EXAM MISTAKE

Students routinely write that with IaaS “the provider manages everything”, which reverses the model. The provider supplies the infrastructure; the customer manages everything built on it, from the operating system up. If your description of IaaS could equally describe SaaS, it cannot score. Anchor every description to who manages what.

Choosing a model: one organisation, three needs

Real organisations rarely pick one model and stop; they match the model to the job. Consider a growing online retailer. For staff email and shared documents it subscribes to Microsoft 365, a SaaS choice, because email is a solved problem and nobody wants to maintain a mail server. For its customer-facing website it uses Azure App Service, a PaaS choice, because its developers want to ship code quickly and let the platform absorb traffic spikes during sales. Its bespoke warehouse-management system, which depends on a specific operating system configuration, runs on Azure Virtual Machines: an IaaS choice, because no higher-level service offers that degree of control. Each decision trades management effort against control, which is exactly the trade-off the syllabus asks you to recognise.

EXAM TIP

Scenario questions reward linking. Do not just name the model: pick out the detail in the scenario that justifies it. “IaaS, because the company needs to choose and configure its own operating system” earns marks; “IaaS” alone may not.

Summary Reference

Model	What the provider supplies	What the customer manages	Typical examples
SaaS	A complete, ready-to-use application plus all underlying infrastructure	Nothing beyond usage and basic settings	Google Workspace, Microsoft 365, Salesforce, Dropbox
PaaS	A development and deployment platform: runtime, operating system, infrastructure	The application code and its configuration	AWS Elastic Beanstalk, Google App Engine, Azure App Service
IaaS	Fundamental resources: virtual servers, storage, networking	Everything from the operating system up, including security and capacity	Amazon EC2, Azure Virtual Machines, Google Compute Engine

Try it on paper

Answer these in writing before you attempt the quiz. Model answers follow each question; mark yourself honestly.

1. Describe what is meant by platform as a service (PaaS). [2 marks]

Model answer: PaaS provides an environment for developing and deploying applications over the internet [1]; the provider manages the underlying servers, storage and operating system, so the customer manages only the application and its configuration [1].

2. A start-up is deciding between SaaS and IaaS for its new customer-booking system. Describe one advantage and one disadvantage of each model in this scenario. [4 marks]

Model answer: SaaS advantage: the system is ready to use with no infrastructure to manage, so the small team can focus on the business [1]. SaaS disadvantage: the software cannot be customised, so a booking system with unusual requirements may not fit [1]. IaaS advantage: full control over the environment, so the system can be built exactly as required [1]. IaaS disadvantage: the start-up must manage the operating system, security and capacity itself, which demands time and expertise it may not have [1].

A 1.1.9 Quiz

1

Which cloud computing service provides access to fundamental computing resources like virtual machines and storage?

- A. ☐ SaaS
- B. ☐ None of the others
- C. ☒ IaaS
- D. ☐ PaaS

2

What is a key characteristic of SaaS (Software as a Service)?

- A. ☐ It is the most complex and flexible cloud service model.
- B. ☐ Users have complete control over the underlying infrastructure.
- C. ☒ Software applications are delivered over the internet, on demand.
- D. ☐ Users need to manage the operating system and software updates.

3

Which cloud service model offers a platform for developing and running applications without managing the infrastructure?

- A. ☒ PaaS
- B. ☐ SaaS
- C. ☐ None of the others
- D. ☐ IaaS

4

In which cloud service model do users have the HIGHEST level of control over the computing environment?

- A. ☐ PaaS
- B. ☒ IaaS
- C. ☐ All models offer the same level of control.
- D. ☐ SaaS

5

Which of the following is an example of a PaaS offering?

- A. ☒ AWS Elastic Beanstalk
- B. ☐ Microsoft Azure Virtual Machines
- C. ☐ Dropbox
- D. ☐ Google Workspace

6

What is a key advantage of using SaaS?

- A. ☐ It provides the most flexibility and customisation options
- B. ☐ It requires extensive technical expertise to manage.
- C. ☐ It is the most expensive cloud service model.
- D. ☒ It offers ease of use and requires minimal maintenance.

7

Which cloud service model is MOST suitable for a small business that needs email and collaboration tools?

- A. ☐ None of the others
- B. ☐ IaaS
- C. ☒ SaaS
- D. ☐ PaaS

8

Which scenario BEST illustrates the use of IaaS?

- A. ☐ A student using Google Docs to write an essay.
- B. ☒ A large company creating a virtualised data center using Amazon EC2.
- C. ☐ A musician streaming music on Spotify
- D. ☐ A developer deploying a web application on Google App Engine.

9

What is a common misconception about cloud computing?

- A. ☐ Cloud computing requires a constant internet connection
- B. ☐ Cloud computing is always more expensive than on-premises solutions.
- C. ☐ Cloud computing is less secure than traditional IT infrastructure.
- D. ☒ Cloud computing is only for large enterprises.

10

Why is it important to understand the different types of cloud services?

- A. ☒ To make informed decisions about choosing the right cloud solution for specific needs and
- B. ☐ It is not important for most computer users.
- C. ☐ To become a cloud computing expert.
- D. ☐ To avoid using cloud services altogether.



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